MALIK JIRASEK

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EDUCATION

Oct 2025 – Present
Master of Science in Physics, University of Tübingen, Germany
Thesis: *"Keldysh-field-theory approach to the quantum Fisher information"*
Supervisor: Igor Lesanovsky

Aug 2020 – Jul 2025
Bachelor of Science in Physics, University of Tübingen, Germany
Final grade: 1.04 / 5.0 (German system; 1.0 = excellent)
Thesis: *"Quantum Sensing beyond the Heisenberg Limit with the Boundary Time Crystal"*
Supervisor: Igor Lesanovsky

RESEARCH EXPERIENCE

Nov 2025 – Present
Master of Science Thesis Research, University of Tübingen, Germany
- Developing a Keldysh-field-theory description of the quantum Fisher information in open quantum systems, with a focus on scalable sensor architectures.
- Results incorporated into a manuscript submitted to *Physical Review Letters* (under review).

Aug 2025 – Sep 2026
Working Student, Bosch eBike Systems, Germany
- Implemented a Machine Learning pipeline for battery cell aging predictions, using BiLSTM networks and TensorFlow.
- Collaborated with the engineering team to integrate the model into the production workflow.

Oct 2024 – Jul 2025
Bachelor of Science Thesis Research, University of Tübingen, Germany
- Demonstrated collective enhancement of sensitivity bounds for measuring optical phases using a boundary time crystal as light source.
- Results incorporated into a paper published in *Physical Review A*.

Aug 2024 – Oct 2024
Research Intern, Humboldt University of Berlin, Germany
- Extended a Fortran-based many-body solver to include high-precision interaction potentials for ultracold-atom tweezer arrays, achieving 10^{-8} relative precision in ground state energies.

PUBLICATIONS

Preprint
Malik Jirasek, Igor Lesanovsky, and Viktoria Noel, "Quantum Fisher information of driven-dissipative systems from Keldysh path integrals". arXiv:2609.26462 [quant-ph].

Published
Malik Jirasek, Igor Lesanovsky, and Albert Cabot, "Boundary time crystal as a light source for collectively enhanced sensing", Phys. Rev. A **114**, 032611(2026).

PRESENTATIONS & CONFERENCES

20 Oct 2026 Invited talk at the Cold Quantum Coffee, Heidelberg

30 Sep 2026 Presented poster at DQUT Conference, Palma de Mallorca

07 Sep 2026 Presented poster at QUANT26 Master School, Dresden

06 Mar 2026 Contributed Talk at DPG Spring Meeting SAMOP, Mainz

TEACHING EXPERIENCE

WS 2024/25	Advanced Quantum Mechanics tutor (University of Tübingen)
SS 2024	Quantum Mechanics tutor (University of Tübingen)
WS 2023/24	Analytical Mechanics tutor (University of Tübingen)
SS 2023	Physics 2 tutor (University of Tübingen)
WS 2022/23	Physics 1 tutor (University of Tübingen)
WS 2021/22 – WS 2024/25	Physics Lab tutor (University of Tübingen)

AWARDS & SCHOLARSHIPS

2021 & 2023	Deutschlandstipendium (merit-based; awarded to ~0.5% of enrolled students)

SKILLS & METHODS

Programming	Scientific programming, high performance computing, machine learning methods
Simulation & Modeling	Stochastic calculus, nonequilibrium statistical mechanics, many-body systems
Soft Skills	Communication between different scientific disciplines, teamwork, problem-solving
Languages	German (native), English (full professional proficiency), French (beginner)

SERVICE & LEADERSHIP

Apr 2027	Co-organizer, Workshop "Dynamics in Strongly Correlated Quantum Open Systems", Tübingen
2026	Student representative, faculty appointment committee (ML in physics)
2022 – 2025	Treasurer & member, physics student council Tübingen